

OS PRACTICAL

PRACTICAL-1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
rishitha@Rishitha: ~/Practical × + ▾
rishitha@Rishitha:~$ cd ~
rishitha@Rishitha:~$ mkdir Practical1
mkdir: Practical1: File exists
rishitha@Rishitha:~$ cd Practical1
rishitha@Rishitha:~/Practical1$ pwd
/home/rishitha/Practical1
rishitha@Rishitha:~/Practical1$ nano practical1.c
rishitha@Rishitha:~/Practical1$ gcc practical1.c -o practical1
rishitha@Rishitha:~/Practical1$ ./practical1
```

```
Enter a Linux command: date

Parent Process
Parent PID = 563

Child Process
Child PID = 564
Mon Aug 17 03:48:10 UTC 2026
Child process completed.
rishitha@Rishitha:~/Practical1$ ./practical1
Enter a Linux command: pwd

Parent Process

Child Process
Parent PID = 576
Child PID = 579
/home/rishitha/Practical1
Child process completed.
rishitha@Rishitha:~/Practical1$ ./practical1
Enter a Linux command: ls

Parent Process
Parent PID = 626

Child Process
Child PID = 627
practical1 practical1.c
Child process completed.
```

```

rishitha@Rishitha:~/Practical1$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              AuthenticAMD
Model name:             AMD Ryzen 5 7530U with Radeon Graphics
CPU family:             25
Model:                  80
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               0
BogoMIPS:               3992.40
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                        r pge mca cmov pat pse36 clflush mmx fxsr sse s
                        se2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtsc
                        p lm constant_tsc rep_good nopl xtopology tsc_r
                        eliable nonstop_tsc cpuid extd_apicid tsc_known
                        _freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_
                        2 movbe popcnt aes xsave avx f16c rdrand hyperv
                        isor lahf_lm cmp_legacy svm cr8_legacy abm sse4
                        a misalignsse 3dnowprefetch osvw topoext perfct
                        r_core ssbd ibrs ibpb stibp vmmcall fsgsbase bm
                        il avx2 smep bmi2 erms invpcid rdseed adx smap
                        clflushopt clwb sha_ni xsaveopt xsavec xgetbv1
                        xsaves clzero xsaveerptr arat npt nrip_save tsc
                        _scale vmcb_clean flushbyasid decodeassists pau
                        sefilter pfthreshold v_vmsave_vmload umip vaes
                        vpcmlmulqdq rdpid fsrm

Virtualization features:
Virtualization:         AMD-V
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    192 KiB (6 instances)
L1i:                    192 KiB (6 instances)
L2:                     3 MiB (6 instances)
L3:                     16 MiB (1 instance)

```

```

NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0-11
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:              Not affected
Indirect target selection: Not affected
Itlb multihit:           Not affected
L1tf:                    Not affected
Mds:                     Not affected
Meltdown:                Not affected
Mmio stale data:         Not affected
Old microcode:           Not affected
Reg file data sampling:  Not affected
Retbleed:                Not affected
Spec rstack overflow:    Vulnerable: Safe RET, no microcode
Spec store bypass:       Mitigation; Speculative Store Bypass disabled v
                          ia prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user
                          pointer sanitization
Spectre v2:              Mitigation; Retpolines; IBPB conditional; IBRS_
                          FW; STIBP always-on; RSB filling; PBRSE-eIBRS N
                          ot affected; BHI Not affected
Srbds:                   Not affected
Tsa:                      Vulnerable: No microcode
Tsx async abort:         Not affected
Vmscape:                 Not affected
rishitha@Rishitha:~/Practical1$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 356.9M  1 disk 
sdb   8:16     0 159.4M  1 disk 
sdc   8:32     0    2G    0 disk [SWAP]
sdd   8:48     0    1T    0 disk /mnt/wslg/distro
/
rishitha@Rishitha:~/Practical1$ ps
PID TTY          TIME CMD
347 pts/0      00:00:00 bash
672 pts/0      00:00:00 ps

```

rishitha@Rishitha:~/Practical1\$ top

top - 03:57:07 up 43 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7612.6 total, 6879.6 free, 575.1 used, 308.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7037.6 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24196	15384	11580	S	0.0	0.2	0:01.42	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.02	init-systemd(Ub
9	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
55	root	19	-1	50396	16864	15568	S	0.0	0.2	0:00.39	systemd-journal
72	systemd+	20	0	22416	14440	11944	S	0.0	0.2	0:00.21	systemd-resolve
94	root	20	0	35252	12348	9188	S	0.0	0.2	0:00.83	systemd-udev
141	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.09	chronyd-starter
145	root	20	0	4472	3104	2860	S	0.0	0.0	0:00.01	cron
148	message+	20	0	8824	5336	4660	S	0.0	0.1	0:00.10	dbus-daemon
168	root	20	0	44096	29796	14660	S	0.0	0.4	0:00.38	networkd-dispat
172	root	20	0	18440	9424	8176	S	0.0	0.1	0:00.17	systemd-logind
192	syslog	20	0	220548	5556	4624	S	0.0	0.1	0:00.12	rsyslogd
244	_chrony	20	0	23804	11228	7116	S	0.0	0.1	0:00.12	chronyd
262	_chrony	20	0	12096	2464	1820	S	0.0	0.0	0:00.01	chronyd
283	root	20	0	123460	32512	17904	S	0.0	0.4	0:00.28	unattended-upgr
289	root	20	0	5492	2744	2556	S	0.0	0.0	0:00.01	agetty
345	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
346	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.35	Relay(347)
347	rishitha	20	0	6268	5528	3860	S	0.0	0.1	0:00.19	bash
348	root	20	0	8648	5548	4704	S	0.0	0.1	0:00.02	login
394	rishitha	20	0	22356	13624	10992	S	0.0	0.2	0:00.27	systemd
396	rishitha	20	0	22920	4108	2104	S	0.0	0.1	0:00.00	(sd-pam)
423	rishitha	20	0	6136	5472	3928	S	0.0	0.1	0:00.08	bash
682	rishitha	20	0	8312	6148	3900	R	0.0	0.1	0:00.36	top